

Assignment 1 extra

Hadeer Ghoneim

Verification Diploma

1.verify functionality of mode when the USE_EN = 1

1.Design add begin and end to if in line 10 to achieve when enable = 0

```
1  module dff(clk, rst, d, q, en);
2  parameter USE_EN = 1;
3  input clk, rst, d, en;
4  output reg q;
5
6  always @(posedge clk) begin
7      if (rst)
8          q <= 0;
9      else
10         if (USE_EN) begin
11             if (en)
12                 q <= d;
13             end
14         else begin
15             q <= d;
16         end
17     end
18
19 endmodule
20
21
```

2. Testbench

```
1  module dff_t1_tb();
2
3      reg  d_tb;
4      logic q_tb, q_expected;
5      reg en;
6      bit clk;
7      bit rst;
8      int error_count;
9      int correct_count;
10
11      dff #(.USE_EN(1)) DUT_DFF_1 (clk, rst, d_tb, q_tb, en);
12
13      dff_GM #(.USE_EN(1)) DUT_DFF_GM_1 (clk, rst, d_tb, q_expected, en);
14
15
16      always begin
17          #5 clk = ~clk;
18      end
19
20      //COUNTER_1
21      task asser_reset ();
22      @(negedge clk);
23          d_tb = 0 ;
24          rst =1;
25          check_fun(0);
26          rst =0;
27      endtask
28
```

```

28
29 //COUNTER_2
30 integer i;
31 initial begin
32     clk = 0;
33     error_count = 0;
34     correct_count = 0;
35
36     asser_reset();
37
38     for(i=0; i<99; i=i+1)begin
39         d_tb = $random;
40         en = $random;
41         check_fun(q_expected);
42     end
43
44     $stop;
45
46 end
47
48 task check_fun(input q_expected);
49 @(negedge clk);
50     if(q_tb != q_expected)
51         error_count = error_count + 1;
52     else
53         correct_count = correct_count + 1;
54 endtask
55
56 endmodule

```

3. Do file

```
1  vlib work
2  vlog dff.v dff_t1_tb.svh +cover -covercells
3  vsim -voptargs=+acc work.dff_t1_tb -cover
4  add wave *
5  coverage save dff_t1_tb.ucdb -onexit -du work.dff
6  run -all|
```

4. Coverage report text file

Coverage Report by instance with details

```
=====
=== Instance: /\dff_t1_tb#DUT_DFF_1
=== Design Unit: work.dff
=====
```

Branch Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Branches	3	3	0	100.00%

=====Branch Details=====

Branch Coverage for instance /\dff_t1_tb#DUT_DFF_1

Line	Item	Count	Source
----	----	----	-----
File dff.v			
-----IF Branch-----			
7		97	Count coming in to IF
7	1	1	if (rst)
11	1	46	if (en)
13	1	50	else

Branch totals: 3 hits of 3 branches = 100.00%

Statement Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Statements	4	4	0	100.00%

=====Statement Details=====

Statement Coverage for instance /\dff_t1_tb#DUT_DFF_1 --

Line	Item	Count	Source
----	----	----	-----
File dff.v			
1			module dff(clk, rst, d, q, en);

=====Statement Details=====

Statement Coverage for instance /\dff_t1_tb#DUT_DFF_1 --

Line	Item	Count	Source
----	----	-----	-----
File dff.v			
1			module dff(clk, rst, d, q, en);
2			parameter USE_EN = 1;
3			input clk, rst, d, en;
4			output reg q;
5			
6	1	97	always @(posedge clk) begin
7			if (rst)
8	1	1	q <= 0;
9			else
10			if(USE_EN)
11			if (en)
12	1	46	q <= d;
13			else
14	1	50	q <= d;

Toggle Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Toggles	10	10	0	100.00%

Toggle Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Toggles	10	10	0	100.00%

=====Toggle Details=====

Toggle Coverage for instance /\dff_t1_tb#DUT_DFF_1 --

Node	1H->0L	0L->1H	"Coverage"

clk	1	1	100.00
d	1	1	100.00
en	1	1	100.00
q	1	1	100.00
rst	1	1	100.00

Total Node Count = 5

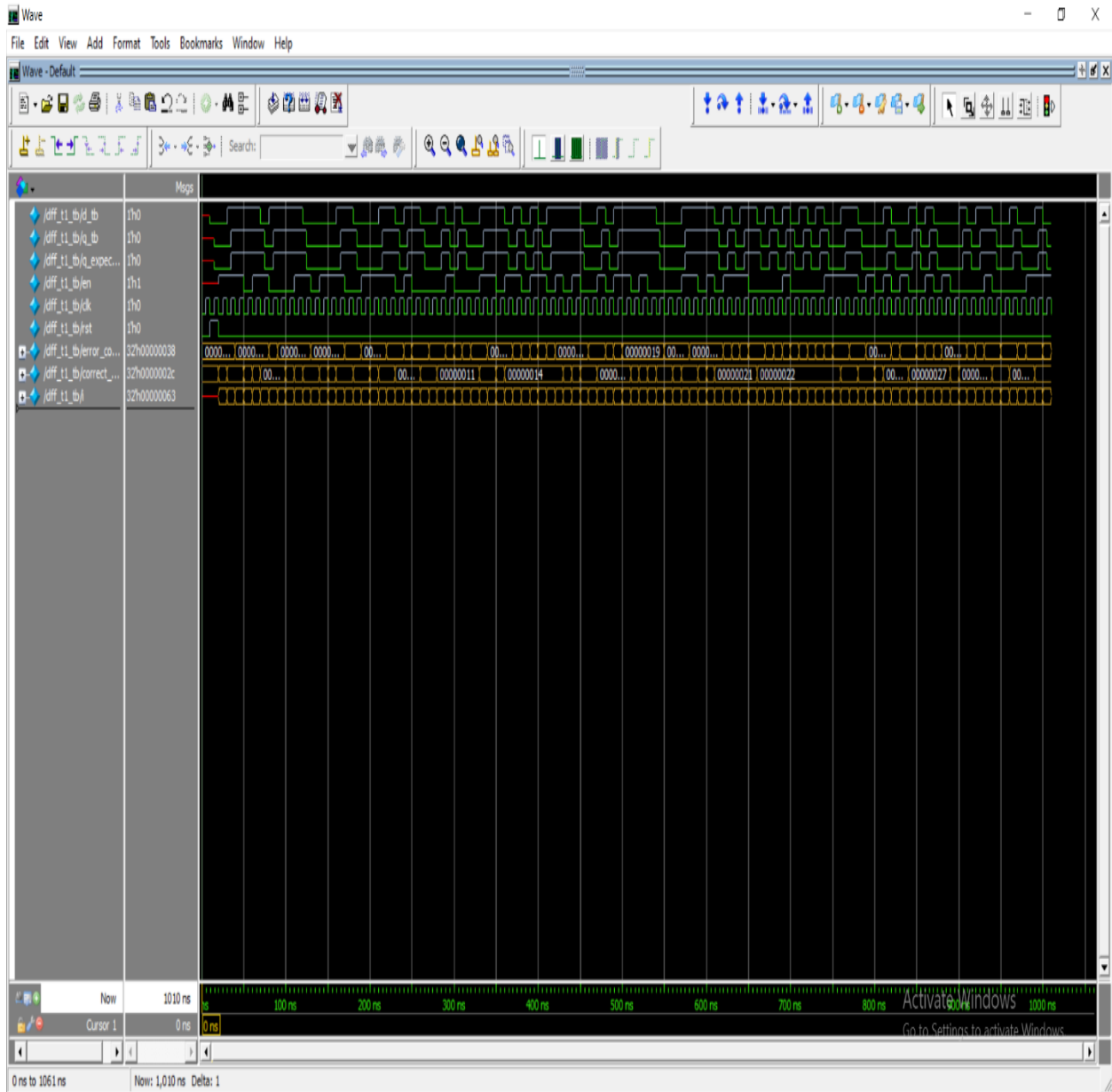
Toggled Node Count = 5

Untoggled Node Count = 0

Toggle Coverage = 100.00% (10 of 10 bins)

Total Coverage By Instance (filtered view): 100.00%

5. QuestaSim waveform snippets



2.verify functionality of mode when the USE_EN = 0

1. same design

2. Testbench

```
1  module dff_t2_tb();
2
3      reg  d_tb;
4      logic q_tb, q_expected;
5      reg en;
6      bit clk;
7      bit rst;
8      int error_count;
9      int correct_count;
10
11      dff #(.USE_EN(0)) DUT_DFF_2 (clk, rst, d_tb, q_tb, en);
12
13      dff_GM #(.USE_EN(0)) DUT_DFF_GM_2 (clk, rst, d_tb, q_expected, en);
14
15
16      always begin
17          #5 clk = ~clk;
18      end
19
20      //COUNTER_1
21      task asser_reset ();
22      @(negedge clk);
23          d_tb = 0 ;
24          rst =1;
25          check_fun(0);
26          rst =0;
27      endtask
28
```

```
28
29 //COUNTER_2
30 integer i;
31 initial begin
32     clk = 0;
33     error_count = 0;
34     correct_count = 0;
35
36     asser_reset();
37
38     for(i=0; i<99; i=i+1)begin
39         d_tb = $random;
40         en = $random;
41         check_fun(q_expected);
42     end
43
44     $stop;
45
46 end
47
48 task check_fun(input q_expected);
49 @(negedge clk);
50     if(q_tb != q_expected)
51         error_count = error_count + 1;
52     else
53         correct_count = correct_count +1;
54 endtask
55
56 endmodule
```

3. Do file

```
1  vlib work
2  vlog dff.v dff_t2_tb.svh +cover -covercells
3  vsim -voptargs=+acc work.dff_t2_tb -cover
4  add wave *
5  coverage save dff_t2_tb.ucdb -onexit -du work.dff
6  run -all
```

4. Coverage report text file

code_cvr_dff_t2 - Notepad

File Edit Format View Help

Coverage Report by instance with details

```
=====
=== Instance: /\dff_t2_tb#DUT_DFF_2
=== Design Unit: work.dff
=====
```

Branch Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Branches	3	3	0	100.00%

```
=====Branch Details=====
```

Branch Coverage for instance /\dff_t2_tb#DUT_DFF_2

Line	Item	Count	Source
----	----	-----	-----
File dff.v			
-----IF Branch-----			
7		90	Count coming in to IF
7	1	1	if (rst)
13	1	45	if (en)
		44	All False Count

Branch totals: 3 hits of 3 branches = 100.00%

Statement Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Statements	3	3	0	100.00%

=====Statement Details=====

Statement Coverage for instance /\dff_t2_tb#DUT_DFF_2 --

Line	Item	Count	Source
----	----	-----	-----
File dff.v			
1			module dff(clk, rst, d, q, en);
2			parameter USE_EN = 0;
3			input clk, rst, d, en;
4			output reg q;
5			
6	1	90	always @(posedge clk) begin
7			if (rst)
8	1	1	q <= 0;
9			else
10			if(USE_EN)
11			q <= d;
12			else
13			if (en)
14	1	45	q <= d;

Toggle Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	-----	-----	-----
Toggles	10	10	0	100.00%

=====Toggle Details=====

Toggle Coverage for instance /\dff_t2_tb#DUT_DFF_2 --

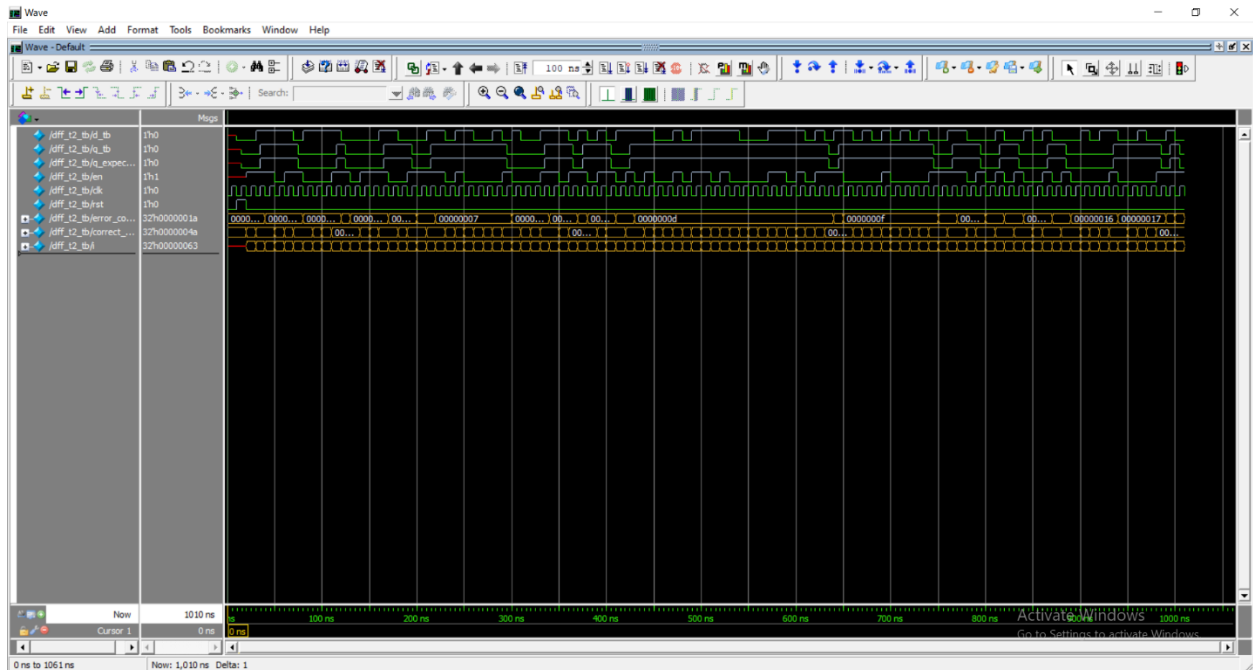
Node	1H->0L	0L->1H	"Coverage"
clk	1	1	100.00
d	1	1	100.00
en	1	1	100.00
q	1	1	100.00
rst	1	1	100.00

Total Node Count = 5
Toggled Node Count = 5
Untoggled Node Count = 0

Toggle Coverage = 100.00% (10 of 10 bins)

Total Coverage By Instance (filtered view): 100.00%

5. QuestaSim waveform snippets



6.dff_merged cover txt report

code_cvr_dff_merge - Notepad

File Edit Format View Help

Coverage Report by instance with details

```
=====
=== Instance: /\work.dff
=== Design Unit: work.dff
=====
```

Branch Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Branches	3	3	0	100.00%

```
=====Branch Details=====
```

Branch Coverage for instance /\work.dff

Line	Item	Count	Source
----	----	----	-----
File dff.v			
-----IF Branch-----			
7		97	Count coming in to IF
7	1	1	if (rst)
11	1	46	q <= d;
13	1	50	if (en)

Branch totals: 3 hits of 3 branches = 100.00%

Statement Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Statements	4	4	0	100.00%

=====Statement Details=====

Statement Coverage for instance /\work.dff --

Line	Item	Count	Source
----	----	----	-----
File dff.v			
1			module dff(clk, rst, d, q, en);
2			parameter USE_EN = 1;
3			input clk, rst, d, en;
4			output reg q;
5			
6	1	97	always @(posedge clk) begin
7			if (rst)
8	1	1	q <= 0;
9			else
10			if (~USE_EN)
11			q <= d;
12	1	46	else
13			if (en)
14	1	50	q <= d;

Toggle Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Toggles	10	10	0	100.00%

- --

Toggle Coverage:

Enabled Coverage	Bins	Hits	Misses	Coverage
-----	----	----	-----	-----
Toggles	10	10	0	100.00%

=====Toggle Details=====

Toggle Coverage for instance /\work.dff --

Node	1H->0L	0L->1H	"Coverage"
-----	-----	-----	-----
clk	2	2	100.00
d	2	2	100.00
en	2	2	100.00
q	2	2	100.00
rst	2	2	100.00

Total Node Count = 5

Toggled Node Count = 5

Untoggled Node Count = 0

Toggle Coverage = 100.00% (10 of 10 bins)

Total Coverage By Instance (filtered view): 100.00%

